University of Cape Town

School of Economics.

Corporate Finance & Equity Valuations Module

ECO4101Z Class Test: 22 June 2023

50 marks Time: 90 minutes

Please read the following instructions carefully

1. Please fill in the attendance slip and complete your personal details on the answer books provided before the start of the test. **No time will be allowed for this at the end of the test.**
2. Cell phones must be switched off for the duration of the test and placed with your other possessions out of sight underneath your chair.
3. Do not open the test paper until instructed to do so. You may not write anything on your test paper or in your answer books before the start of the test or once you have been instructed to stop writing.
4. All answers must be written in black or blue ink including your workings
5. **No questions will be answered by invigilators during the test.** Please make whatever assumptions you deem appropriate and clearly state these in your answer.
6. Only none programmable calculators are permitted in this test.
7. Answer all questions, and show all your workings where calculations are required.
8. This test paper consists of 5 pages including cover page and 5 questions numbered **Question 1 to Question 5**

Question 1 **(15 marks)**

Assume that Zion Ltd's management is considering a handful of projects for next year. The Investment Analysts in the company were busy evaluating some of the potential projects in the past week for possible inclusion in the current portfolio of assets. Below is a tabular presentation of the projects under consideration.

Project	IRR	Capital required
A	14.5%	R350 000
B	13.5%	R250 000
C	11.5%	R180 000
D	12.8%	R400 000

Further information shows that management is considering financing future projects as follows. (1). Target capital structure of 40% (E), 40% (D) and 20% (E) using Equity, Debentures and preference shares respectively.

$DY = \frac{D}{P_0}$

(2). Use preference shares that have a dividend yield (DY) of 12% and a current price of R10. New preference shares attract 5% flotation (FC) costs.

(3). Management assumes that the market-related interest rates for debentures with similar characteristics will be 10% (i) and the company's marginal tax rate is 28% (t).

(4). Furthermore, additional information that was gathered to assist in the calculation of the cost of equity shows that: A return on 1 year treasury bill is 6%, Return on the 10-year government bond is 7%, company Beta (β) of 1.2, risk premium ($r_m - r_f$) of 9%.

$k_e = r_f + \beta_1 (r_m - r_f)$

Required

(a) State the projects that should be accepted and provide a reason why those projects should be included in the portfolio next year. <u>**Show all your calculations**</u> (11 marks)

(b) Given the above information, how much is the budget (1 mark)

(c) State the amounts that must be raised from each source. (3 marks)

a) $WACC = kd(1-t)\,wd + k_{os}\,w_{os} + k_p\,w_p$

$= 0{,}1 \times .72\,(.4) + 0{,}18\,(.4) + 0{,}1206\,(.2)$

$= 12{,}48\%$ 0,16?

$DY = \frac{D}{P_0}$ $\quad 12\% = \frac{D}{10}$

$D = R1{,}20$

$kp = \frac{D_1}{P_0 - F_c} = \frac{1{,}20}{10 - 5\%} = 0{,}12$

$ke = \frac{D_1}{P_0} + g = \frac{1{,}2}{10} + 6\% = 0{,}18$

Projects accepted : A, B, D

reason : their return (IRR) is higher than the cost (WACC) of doing them

IRR > WACC

b) Budget = 350k + 250k + 400k = R1M

c) Equity : R400k

Debentures : R 400k

Pref. shares : R 200k

Question 2

a) 1- Residual approach × a div. has an effect on value of comp - ∴ relevant

2- Determine - part of budget × div. has no effect on value of comp ∴ irrelavant

b) 1- Future projects (investment opportunities)

2- RE RRR

3- Creditors Target debt ratio

4- The signalling effect

c) $WACC = kd(1-t)\,wd + k_e\,W_e$

$= .11\,(.75) + .12\,(.25) = 11{,}25\%$

Accept project A & C ∴ Budget = 64 + 40k = R104k

d) 1- expected NPAT investment decision (TAT)

2- signalling effect financing decision (EM)

3- distribution decision - dividend decision

Question 2 **(15 marks)**

(a) Name two schools of thought that exist regarding the distribution of dividends **(2 marks).**

(b) State any four factors that a company should take into account in their dividend policy before distributing a dividend. **(4 marks)**

(c) Assume that Bonakale Ltd is currently evaluating its ordinary dividend policy and that in the previous years Bonakale Ltd has followed a stable dividend policy of D_0 R0.50 per share plus bonus. Assume that the bonus is based on any residual income available. Further assume that the company maintains a target capital structure of 1 E 25% Equity and 3 D 75% Debt. In addition, assume that the cost of capital is k_e 12% and that the after tax cost of debt $k_d(1-t)$ is 11%.

Required

Calculate the residual income available to shareholders after taking into account the information below:

Bonakale Ltd is expecting NPAT R40 million to be available as a net profit after tax next year, of which R36 million will be available in cash for investment purposes. Assume that the company has the following investment opportunities:

Project	Cost	Expected return
A	R64 000 000	15%
B	R50 000 000	10%
C	R40 000 000	13%

Note: your answer should show all the calculations including reasons **(6 marks)**

(d). The residual approach suggests that dividends are a passive variable. A dividend is what is left over after the investment and financing decisions have been made. However, to apply the residual approach, a company should identify three (3) important factors, name these factors. **(3 marks)**

3a) 1 - k_d (i)
2 - tax shield
3 -

b) Pecking order theory:

4a) $P_r = \frac{45}{10} = R4{,}50$

b) $V_F = 90m\,(5) + 45m = R495m$

c) $\frac{495}{100} = R4{,}95$

d) $V_r = R4{,}95 - 4{,}50 = R0{,}45$

5a) $W_d = 43\%$
$W_e = 57\%$

$1 : \frac{1}{42{,}8572}$

$\frac{1}{x} = 0{,}42$

b) $WACC = k_d(1-t)\,W_d + k_e\,W_e$
$= 0.1\,(1-0{,}28)\,.43 + .25\,(.57)$
$= 17{,}346\%$

Question 3 **(5 marks)**

(a) Capital structure plays a significant role in the calculation of the WACC. However, the debt policy is dependent on a range of factors. Name three (3) of these factors that would

CS OP

influence a company's capital structure and debt policy. **(3 marks)**

CS

(b) Briefly describe the perking order theory of capital structure **(2 marks)**

Question 4 **(5 marks)**

90m P_0

Tholakale Ltd currently has 90 million shares in issue. The current share price is R5 and plans

10m

to raise R45 million through a rights issue of 10 million shares.

Required

(a) What is the subscription price of each rights issue **(1 mark)**

(b) Calculate the value of the company after the rights issue **(1 marks)**

(c) Calculate the theoretical value of a share **(1 mark)**

(d) Calculate the value of a right **(2 marks)**

Question 5 **(10 Marks)**

Azua Ltd is in the business of chicken production at the Sun Valley estate. The company would like to raise R600 million next year to finance an outlet in the city of Cape Town. Azua currently

D:E .428572 : 1

has a debt to equity ratio of 42.8572%, which it wishes to maintain as its target capital structure when raising future capital. The current financial year end shows that Azua Ltd.'s

NPAT D

reported earnings after tax is R300 million. The company expects to pay a dividend of R80 million from its expected earnings after tax next year. The company has a pay-out policy of

payout

20%. Furthermore, it is the company's policy to exhaust its internally generated earnings available to finance its future capital expenditure projects before issuing any new ordinary shares.

k_{re} k_e $k_d + FC$ FC t

The cost of retained earnings is 22%, the cost of issuing new shares is 25%, while the before tax cost of debt including flotation costs is 15%. Flotation costs on issuing debt are 5%. Tax rate is 28%.

Required

(a) Calculate percentage capital structure weights before raising new capital, your answer must be nearest 1 percent. (E.g. 10%, not 10.25%). **(2 marks)**

(b) Calculate percentage capital structure weights required to raise new capital.**(6 marks)**

(c) Calculate the cost of capital (WACC). **(2 marks)**